



Multiphysics modeling of porous CRUD deposits in nuclear reactors



M.P. Short^{a,*}, D. Hussey^b, B.K. Kendrick^d, T.M. Besmann^c, C.R. Stanek^d, S. Yip^a

^aMassachusetts Institute of Technology, 77 Massachusetts Ave., Cambridge, MA 02139, United States

^bElectric Power Research Institute, 3420 Hillview Avenue, Palo Alto, CA 94304, United States

^cOak Ridge National Laboratory, P.O. Box 2008, MS6063, Oak Ridge, TN 37831-6063, United States

^dLos Alamos National Laboratory, P.O. Box 1663, Los Alamos, NM 87545, United States

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ABSTRACT

The formation of porous CRUD deposits on nuclear reactor fuel rods, a longstanding problem in the operation of pressurized water reactors (PWRs), is a significant challenge to science-based multiscale modeling and simulation. While existing, published studies have focused on individual or loosely coupled processes, such as heat transfer, fluid flow, and compound dissolution/precipitation, none have addressed their coupled effects sufficiently to enable a comprehensive, scientific understanding of CRUD. Here we present the formulation and results of a model, MAMBA-BDM, which begins to incorporate mechanistic details in describing CRUD in PWRs. CRUD is treated as a chemical deposition process in an environment of variable concentration, an arbitrary level of heating, and a complex fractal-based flow geometry. We present results on spatial distributions of temperature, pressure, velocity, and concentration that give insight into the interplay between these physical properties and geometrical parameters. We show the role of heat convection which has not been discussed previously. Furthermore, we suggest that the assumption of liquid saturation in the CRUD deserves scrutiny, as a result of our attempt to determine an effective CRUD thermal conductivity.

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1. Introduction

One of the main challenges in achieving higher burnup and power uprates in light water reactors (LWRs) is the problem of fouling of reactor fuel rods [1]. This phenomenon, known as Chalk River Unidentified Deposits (CRUD) after the location of their discovery, is a major concern in fuel reliability and reactor operation [2,3], with further consequences concerning safety under accident conditions. In existing CRUD models, simplifications such as infinitely soluble species, one-dimensional analysis, constant material properties, liquid saturation, and decoupled physical phenomena had been introduced. Although useful at the time, it is desirable to relax these assumptions by a formulation that is more science based and also scalable to massively parallel, high performance computation.

In this paper, a new mesoscale CRUD model, known as MAMBA-BDM¹ (for Boron Deposition Model), is formulated and implemented in a physics simulation framework, MOOSE (Multiphysics Object

Oriented Simulation Environment), which is dimension and geometry agnostic [4]. The multiphysics nature of CRUD formation and growth is treated explicitly by allowing heat transport to have both conductive and convective contributions. The calculation of temperature, fluid pressure, fluid velocity, spatially-dependent material properties, soluble boron concentration, and regions of boron oxide precipitation in the CRUD is performed simultaneously. A control volume formulation consisting of one boiling chimney and its surrounding CRUD allows for the treatment of CRUD with a regular spacing of boiling chimneys.

In Section 2, we provide a brief background on the importance of CRUD in fuel performance, a brief survey of previous models, and the solubility data for boron compounds. The MAMBA-BDM code is described in Section 3. Results are reported in Section 4 in three groups. First, spatial distributions of temperature, pressure, velocity, and boron concentration show peak values that give insight into the complex mechanisms underlying CRUD formation and growth. Secondly, the significance of heat convection and its dependence on chimney spacing (CRUD density) is demonstrated. Lastly, the assumption of liquid saturation is called into question by noting an inconsistency in the estimate of an effective thermal conductivity of CRUD when compared to experiments. Implications for future CRUD models and other related branches of engineering are briefly given in Section 5, while concluding remarks are given in Section 6.

* Corresponding author.

E-mail addresses: hereiam@mit.edu (M.P. Short), dhussey@epri.com (D. Hussey), bkendric@lanl.gov (B.K. Kendrick), besmann@ornl.gov (T.M. Besmann), stanek@lanl.gov (C.R. Stanek), syip@mit.edu (S. Yip).

¹ MAMBA (MPO Advanced Model for Boron Analysis) is a related engineering-scale CRUD deposition model being developed by the MPO (Materials Performance and Optimization) focus area within CASL.

2. Background

2.1. CRUD's effects on fuel performance

There are three main concerns in reactor operation attributed to CRUD. First is a neutronic effect due to the presence of boron build-up in CRUD, known as a CRUD induced power shift (CIPS), which is a downwards axial shift in the power distribution. The second effect is CRUD-induced localized corrosion (CILC), where degraded heat transfer due to CRUD accelerates corrosion processes. A third concern is the increased radioactivity of the fuel rod, because of neutron-activated isotopes present in the CRUD [2,5]. The first two of these effects on fuel performance is discussed in the following subsections, as they can be addressed by the results of MAM-BA-BDM.

CIPS – CRUD tends to grow on the upper spans of PWR fuel rods [2], due to the postulated mechanisms of nucleation associated with sub-cooled boiling [6,7]. A schematic of the proposed CRUD nucleation process is shown in Fig. 1. As bubbles form, supersaturated solutes which may not have fully dissolved are pushed out of solution, so that when the bubbles collapse, the solutes are left behind as porous deposits. As these deposits build up, boric acid present in PWR water (which may reach 1200 ppm by mass at the start of a fuel cycle [2]) enters the CRUD. Once the temperature increases and sub-cooled boiling begins, these solutes can precipitate inside the pores of the CRUD, loading the upper fuel spans with boron. This leads directly to CIPS, where the boron trapped in the porous CRUD locally depresses the neutron flux. Boron deposition means a loss of controllability in the event of an accident, because the emergency control rods enter from the top of the reactor. During CIPS, the control rod worth decreases, thus CIPS can lead to mandatory derating of a plant for a given fuel cycle.

CILC – The presence of CRUD can cause both the cladding temperature and the solute concentrations at the cladding surface to increase under certain conditions, like thick or low-porosity CRUD [2,3]. Thick CRUD therefore gives rise to large temperature gradients at the CRUD-clad interface, and consequently accelerates corrosion. Localized 'hotspots' can provide initiation sites for more rapid corrosion, and produce "CILC leakers," at locations underneath CRUD where fuel rod penetration can take place.

2.2. Previous CRUD models

Heat transfer across the CRUD deposit was described in the Cohen model by modeling the heated surface in the form of a 1D boiling chimney [8]. In this model, heat flows from the clad surface into the CRUD shell, and out across the chimney surface where boiling is assumed to occur. The CRUD is treated as a porous deposit with prescribed porosity, pore size and tortuosity. The steam

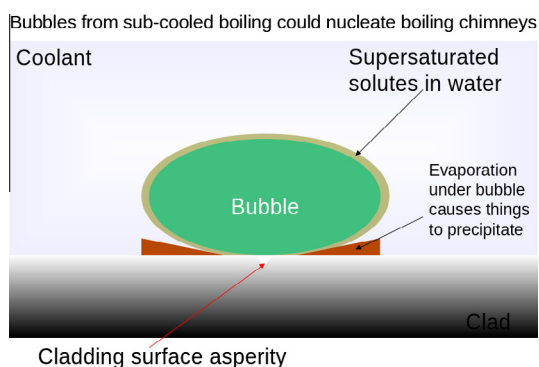


Fig. 1. Schematic showing how CRUD has been proposed to nucleate during sub-cooled boiling.

evaporation rate is governed by a 1D temperature profile along the chimney. This description did not allow for dryout regions in the porous deposit, nor does it consider property variations of parameters such as porosity or tortuosity perpendicular to the chimney axis.

A 2D extension of the Cohen model has been implemented by Pan et al. [9] as a CRUD Chemistry Model (CCM). Chemical concentration effects were also included by treating the concentration of an "infinitely soluble" solute (in this case, $\text{H}_3\text{BO}_3(\text{aq})$) as a result of wick boiling, thus accounting for changes in boiling surface and fluid transport due to porosity and tortuosity. However, the infinite solubility assumption could give rise to unrealistic soluble concentration levels. In addition, the assumption that the boiling surface is at the boiling chimney may not be valid, as limits in solute concentration would impose strict limits on boiling point elevation compared to those in this model.

In the model by Pan et al. CCM equations for spatially-dependent concentrations are formulated for the local region surrounding the boiling chimney, then solved numerically on a roughly 50×50 dimensionless mesh, scaled by the size of the computational cell and the CRUD thickness. In comparing the 1D and 2D results, Pan et al. show that the Cohen model appears to overestimate the concentration of boric acid, as it lies close to the maximum of the 2D calculation. Pan's model, however, could give an overestimate of boron concentration because of the assumption of infinite solubility, while not treating precipitation could underestimate the boron concentration. In actuality, dryout will limit the amount of soluble boron-bearing species that can be formed. Furthermore, Pan et al. demonstrate the effect of varying the porosity on the maximum concentration factor in CRUD, where a porosity change of 20% can change the boron concentration by an order of magnitude. The model assumes a constant porosity at steady state, so allowing porosity to vary with distance and time will add a dimension of realism to the model.

The CCM by Henshaw et al. [10] considers the concentration of multiple boron species coupled with a detailed chemistry model, a thermal transport model, radiolysis chemistry, depletion of boron by neutron absorption, lithium monoborate precipitation, and wick boiling. It is based on the 1D Cohen model, with computational simplicity suitable for a desktop computer. Parameters such as CRUD pore size and tortuosity are taken as constants, while concentrations of soluble and gaseous species can vary along the axis of the boiling chimney. CRUD thermal conductivity is calculated using the Maxwell relation [11]. The chemistry model is highly detailed, including forty equations for the radiolysis of water alone. These equations and their rate constants were taken from the work of Elliot et al. [12]. The resulting description is capable of showing an elevated pH near the cladding wall, which in turn leads to instability of zirconia. It should be noted that the Henshaw model uses a single chimney size and spacing.

The CCM by Haq et al. [13] is a 2D description of heat flux and solute concentration assuming boiling chimneys of fixed size and spacing. This CCM has shortcomings similar to the previous models, with the exception of the Henshaw model, in that it addresses only the concentration of one boron-bearing species, H_3BO_3 . On the other hand, it does treat the coupling of heat transfer, solute concentration, and coolant velocity more thoroughly than some previous models. Results illustrating the effects of coupling show a clear deviation from uncoupled models. The model does account for the increase in saturation temperature due to solute concentration in the case of only one solute.

2.3. Boron compound solubility data

It is known Li–B compounds exhibit retrograde solubility above a certain temperature [14]. This means that beyond a transition

temperature their solubilities decrease with increasing temperature. The data for $\text{Li}_2\text{B}_4\text{O}_7$ stop well below the temperatures present in CRUD [14]. One way to extend the data range could be first-principles electronic structure calculations of solubility and transport in pores similar to those found in CRUD [15].

3. The MAMBA-BDM framework

The present boron deposition model in 2D form treats a single boiling chimney and its surrounding revolved volume of CRUD as its control volume, which is its smallest repeatable unit. This choice of control volume is based on photographs of CRUD scrapes from PWRs [2]. The single boiling chimney cell becomes a modular input to a larger length scale program external to MAMBA-BDM. This allows the modeling of an assembly of computational cells, each with its own boiling chimney parameters, such as coolant temperature, cladding heat flux, and neutron flux, to study the boron concentration variation across an entire fuel rod.

The following description of the MAMBA-BDM framework, key results, and discussions comparing MAMBA-BDM's initial results with previous models and other fields of engineering will be given in the sections below. The sections focus on the differences between previous models and those in MAMBA-BDM.

3.1. CRUD material models

3.1.1. Fractal material assumptions

CRUD is treated as a statistically self-similar porous solid across multiple length scales. This assumption is based on observations of the oxides that comprise CRUD, which are corrosion products comprised partially from corrosion products stemming from the reactor's internal surfaces. Example SEM micrographs of these CRUD constituents, which partially form the CRUD skeleton, are shown in the work of Combrade et al. [16] and Neves et al. [17].

A multitude of material properties can be computed from a few fractalline geometry parameters. These include the pore size distribution, the porosity, the pore free area dimension (D_f), and the tortuous capillarity dimension (D_T) [18]. D_f accounts for the pore size distribution in fractal geometry space, while D_T accounts for the tortuous fluid paths created by the surfaces of a porous solid approximated by a fractal. These two fractal dimensions are calculated as given by [18] as follows:

$$D_f = d_e - \frac{\ln(\epsilon)}{\ln\left(\frac{\lambda_{\min}}{\lambda_{\max}}\right)} \quad (1)$$

$$D_T = 1 + \frac{\ln(\tau)}{\ln\left(\frac{L}{\lambda_{\min}}\right)} \quad (2)$$

where d_e is the dimensionality of the simulation (2 for 2D, 3 for 3D), ϵ is the spatially dependent CRUD porosity, λ is the CRUD pore diameter, and L is a characteristic length of the system, defined as follows in [18]:

$$L = \lambda_{\max} \sqrt{\frac{\pi}{4} \frac{D_f}{2 - D_f} \frac{(1 - \epsilon)}{\epsilon}} \quad (3)$$

The pore size diameters used in this study are set at $\lambda_{\min} = 0.25 \mu\text{m}$, $\lambda_{\max} = 0.50 \mu\text{m}$, $\lambda_{\text{avg}} = \frac{\lambda_{\min} + \lambda_{\max}}{2}$, based on analysis of SEM micrographs of CRUD [2,19]. More detailed transmission electron microscope (TEM) examinations will be necessary to refine these pore size distributions, as it is expected that the self-similarity of CRUD will extend beyond the length scales of the pore sizes used in the current simulations. In addition, TEM may be necessary to resolve small pores not resolvable in a scanning electron microscope

(SEM), as well as to perform crystallography studies on individual CRUD particles. In MAMBA-BDM, the CRUD/fluid thermal conductivity distribution, the Darcy flow permeability distribution, and the CRUD pore tortuosity distribution are computed from the above fractal dimensions. The tortuosity is computed as follows, directly from the porosity [18]:

$$\tau = \frac{1}{2} \left[1 + \frac{1}{2} \sqrt{1 - \epsilon} + \sqrt{1 + \epsilon} \frac{\sqrt{\left(\frac{1}{\sqrt{1 - \epsilon}} - 1\right)^2 + \frac{1}{4}}}{1 - \sqrt{1 - \epsilon}} \right] \quad (4)$$

These fractal dimensions can be directly measured through pore size distributions using the method by Pitchumani and Ramakrishnan [20] to confirm the fractal material models in MAMBA-BDM.

3.1.2. Property correlations

A number of property correlations are used to describe physical and chemical property variations in the CRUD, rather than using fixed values. Correlations for the solubilities of boron and lithium bearing species are implemented, but currently the precipitation of just boron oxide (B_2O_3) by dehydration of boric acid (H_3BO_3), and then metaboric acid (HBO_2), is implemented in MAMBA-BDM. This is due to a lack of reliable data for the solubilities of key phases, such as lithium tetraborate, at PWR-relevant temperatures [14]. The density, thermal conductivity, heat capacity, and viscosity of water as functions of pressure and temperature are taken from [21], and are specified in later sections. The saturation temperature of water with high amounts of boric acid was taken from Deshon [2]. A polynomial function was fit to the data, with boric acid concentrations scaled to mole fractions in the fluid inside the CRUD:

$$T = 11,453.3C_{\text{BO}_3}^3 - 249.862C_{\text{BO}_3}^2 + 274.056C_{\text{BO}_3} + 618.068 \quad (5)$$

where T is the temperature in Kelvin and C_{BO_3} is the concentration of soluble boric acid.

3.1.3. CRUD thermal conductivity

Thermal conductivities of solid phases in the CRUD, including Ni [22], NiO [23], Fe_3O_4 [24], and ZrO_2 (polycrystalline) [23], are also temperature dependent. The overall thermal conductivity of the solid parts of the CRUD (k_s) is defined as a volume-fraction average of individual components' thermal conductivities:

$$k_s = \prod_{i=1}^n [k_{s,i} v_{f,i}] \quad (6)$$

where $k_{s,i}$ is the thermal conductivity of one phase from Table 1, $v_{f,i}$ is its corresponding volume fraction, and n is the number of solid components considered. The thermal conductivity of water (k_f) is taken from an MIT report describing correlations for water at relevant temperatures and pressures [21]:

$$k_f = 0.686 + 7.3 \times 10^{-10}P + 5.87 \times 10^{-6}(T - 415)^2 \quad (7)$$

where P is the pressure in Pa and T is the temperature in degrees Kelvin. The overall CRUD thermal conductivity is assumed to have equal contributions from parallel and series conductances [25], and the CRUD is assumed to have a spatially uniform microstructure:

$$k_{\text{parallel}} = \frac{(2 - D_f) \epsilon \lambda_{\max}^{D_f-1} \left[1 - \left(\frac{\lambda_{\min}}{\lambda_{\max}} \right)^{D_f-D_f+1} \right]}{t_{\text{CRUD}}^{D_f-1} (D_T - D_f + 1) \left[1 - \left(\frac{\lambda_{\min}}{\lambda_{\max}} \right)^{2-D_f} \right]} k_f + (1 - \epsilon) k_s \quad (8)$$

$$k_{\text{series}} = \frac{1}{\frac{\epsilon}{k_f} + \frac{1-\epsilon}{k_s}} \quad (9)$$

Table 1

Thermal conductivities of solid phases found in the CRUD, with assumed volume fractions for this study.

Species	k ($\frac{W}{mK}$)	Solid volume fraction in results
Ni	$703.56 - 83.297 \times \ln(T + 1518.17)$ [22]	0
NiO	$-5.602 \times 10^{-9} \cdot T^3 + 2.435 \times 10^{-5} \cdot T^2 - 0.03543 \cdot T + 21.658$ [23]	0.15
Fe ₃ O ₄	$4.23 - T \cdot (1.37 \times 10^{-3})$ [24]	0.1
NiFe ₂ O ₄	8 (currently unknown)	0.75
ZrO ₂ (polycrystalline)	$3.791 \times 10^{-4} \cdot T + 1.748$ [23]	0

the latter of which is the inverse of series thermal resistances: The effective series thermal resistance of the CRUD is defined as follows:

$$R_{series} = \varepsilon R_f + (1 - \varepsilon) R_s \quad (10)$$

Finally, a parameter ξ can be defined as the series/parallel character of the CRUD to combine Eqs. (9) and (8) using the methodology of Eqs. (10) and (9):

$$k_{CRUD} = \frac{1}{\frac{\xi}{k_{series}} + \frac{1-\xi}{k_{parallel}}} \quad (11)$$

Here, $\xi = 0$ refers to all capillaries in the direction of flow, and therefore parallel resistances, while $\xi = 1$ refers to all perpendicular capillaries, and therefore series resistances. MAMBA-BDM assumes $\xi = 0.5$. This assumption stems from micrographs of CRUD, which show no directionality to the pore structure in any direction [2,19].

3.1.4. Flow permeability

The flow permeability (κ) is calculated using the assumption that the inner structure of the CRUD has a fractalline nature. This idea has been suggested under similar conditions for porous deposits formed by oxides [26,27]. Analysis of the permeability of a fractalline or fractal-like solid involves calculating its porosity, its tortuosity, and the two fractal dimensions from Eqs. (21), (26), (30a) and (65) in [18]:

$$\kappa = \frac{(\pi D_f)^{\frac{(1-D_f)}{2}} [4(2-D_f)]^{\frac{(1+D_f)}{2}}}{128(3+D_f-D_f)} \left(\frac{\epsilon}{1-\epsilon} \right)^{\frac{(1+D_f)}{2}} \lambda_{max}^2 \quad (12)$$

A larger D_f corresponds to the presence of more, smaller pores, and a correspondingly decreasing permeability [20]. A higher D_f corresponds to a higher flow resistance, due to more tortuous and convoluted capillaries.

3.2. Heat transfer modeling

In the 2D case, the temperature distribution T inside the CRUD around one boiling chimney is assumed to be azimuthally symmetric, so that a 2D slice is all that is necessary to model it. The governing equation is of the conduction–convection form:

$$-\nabla \cdot (k_{CRUD} \nabla T) + \rho c_p \tau \left(\frac{\kappa}{\mu \epsilon} \nabla P \right) \nabla T = 0 \quad (13)$$

where k_{CRUD} is the calculated fractalline thermal conductivity of the CRUD–fluid mixture, ρ is the average CRUD density, c_p is the specific heat capacity of the CRUD, given as follows [21]:

$$c_{p_{H_2O}} = 4028 + \frac{128.8}{\left(1 - \frac{T}{650}\right)} + \frac{4.674}{\left(1 - \frac{T}{650}\right)^2} \quad (14)$$

μ is the fluid viscosity in the CRUD, given as follows [21]:

$$\eta_{H_2O} = \frac{25.3}{-85800 + 91 \cdot T + T^2} \quad (15)$$

and P is the pressure distribution, found below. The quantity τ accounts for convection along highly tortuous pore networks. To solve

for T everywhere, boundary conditions are imposed on the control volume to arrive at a unique solution as follows:

$$-k_{CRUD} \frac{\partial T}{\partial n} = \dot{Q}_{clad}; \quad \Gamma = \Gamma_{clad} \quad (\text{Neumann}) \quad (16)$$

$$T = T_{sat}; \quad \Gamma = \Gamma_{chimney} \quad (\text{Dirichlet}) \quad (17)$$

$$-k_{CRUD} \frac{\partial T}{\partial n} = h_{conv}(T - T_{coolant}); \quad \Gamma = \Gamma_{coolant} \quad (\text{Neumann}) \quad (18)$$

$$\frac{\partial T}{\partial n} = 0; \quad \Gamma = \Gamma_{symmetric} \quad (\text{Symmetric}) \quad (19)$$

where Γ refers to a specific boundary, $\vec{n}$ is a vector normal to a boundary, $\dot{Q}_{clad}$ is an imposed cladding heat flux, and h_{conv} is an imposed heat transfer coefficient at the CRUD–coolant boundary.

3.2.1. Péclet number

A useful metric to determine the relative contributions of heat conduction through the CRUD and heat convection through the fluid is to compute the smeared Péclet number of the CRUD. The Péclet number is a dimensionless constant that shows the ratio of conduction to convection,

$$Pe = \frac{\Delta H_{convection}}{\Delta H_{conduction}} = \frac{\dot{m} c_p \Delta T}{\frac{k_{CRUD}}{\delta} A \Delta T} \quad (20)$$

where ΔH is the enthalpy change, A is the total cross sectional area for flow through the pores, and δ is the characteristic length of the system. Here, the relevant Péclet number measures the total enthalpy transferred by convection versus conduction throughout the entire modeled mass of CRUD, so δ is equal to the thickness of the CRUD (t_{CRUD}). The mass flux is defined as

$$\dot{m} = \rho \vec{v} A \epsilon \quad (21)$$

where $\vec{v}$ is the true fluid pore velocity, defined as follows:

$$\vec{v} = \frac{-\kappa}{\mu \epsilon} \nabla P \quad (22)$$

Porosity is included in order to account for the reduction in cross sectional area, and it varies locally with CRUD microstructure and precipitation of B- and Li-bearing species. Therefore, after canceling terms, the Péclet number becomes

$$Pe = \frac{\rho \left(\frac{\kappa}{\mu} \nabla P \right) c_p t_{CRUD}}{k_{CRUD}} \quad (23)$$

which can be computed locally as an indication of where heat convection becomes non-negligible.

3.3. Coolant flow modeling

The simulation space, boundary definitions, and governing equations for the velocity distribution v (found by solving for the pressure distribution P) are the same as those used in the temperature distribution, with the exception of boundary conditions and one parameter. The equation is again Laplace's formula, with the

thermal conductivity replaced by a flow permeability (κ) in accordance with Darcy's law [28]:

$$-\nabla \cdot \left(\frac{\kappa}{\mu \epsilon} \nabla P \right) = 0 \quad (24)$$

Boundary conditions are specified as follows:

$$\frac{\partial P}{\partial n} = 0; \quad \Gamma = \Gamma_{\text{clad}} \quad (25)$$

$$\frac{\kappa}{\mu} \frac{\partial P}{\partial n} = \frac{-k_{\text{CRUD}}}{\rho h_{\text{fg}}} \cdot \frac{\partial T}{\partial n}; \quad \Gamma = \Gamma_{\text{chimney}} \quad (26)$$

$$P = P_{\text{coolant}}; \quad \Gamma = \Gamma_{\text{coolant}} \quad (27)$$

$$\frac{\partial P}{\partial n} = 0; \quad \Gamma = \Gamma_{\text{symmetric}} \quad (28)$$

where h_{fg} is the enthalpy of vaporization of saturated water, used in the wick boiling assumption at the CRUD-chimney boundary.

3.4. CRUD chemistry modeling

Based on EPRI reports [2,3] which reviewed CRUD scrapes from plants operating in the US, a CCM has been developed by Henshaw et al. [10]. This included investigations by the authors using the FactSage thermodynamic database, in addition to detailed theoretical and experimental studies of polyborate complex equilibria [29]. Boric acid concentration build-up due to fluid convection is assumed to continue until solubility limit (dependent on temperature and total species concentration)) is reached. One species, boric acid, is explicitly modeled as the primary variable in a partial differential equation, and other species' concentrations (auxiliary variables) are based on this value through ordinary differential equations or algebraic calculations. The overall equation for each soluble chemical species is:

$$[(\vec{v} \cdot \vec{n}) - \nabla \tau D_i] \nabla C_i + \sum [C_{i,\text{source}} - C_{i,\text{sink}}] = 0 \quad (29)$$

where C_i is the concentration of a given species i , and D_i is its temperature-dependent diffusion coefficient in water. Boundary conditions are defined as follows:

$$\frac{\partial C_i}{\partial n} = 0; \quad \Gamma = \Gamma_{\text{clad}} \quad (30)$$

$$-\tau D_i \nabla C_i + (\vec{v} \cdot \vec{n}) C_i = 0; \quad \Gamma = \Gamma_{\text{chimney}} \quad (31)$$

$$C_i = C_{i,\text{coolant}}; \quad \Gamma = \Gamma_{\text{coolant}} \quad (32)$$

$$\frac{\partial C_i}{\partial n} = 0; \quad \Gamma = \Gamma_{\text{symmetric}} \quad (33)$$

The diffusion coefficient is scaled up using the pore tortuosity distribution τ , as this gives an effective diffusive path length along a highly tortuous network of pores. Sources of each species may come from chemical reactions, both aqueous and precipitative, and sinks come from both chemical reactions and burning of boron by the (n, α) reaction from an imposed neutron flux. One ion, BO_3^{-3} , and one solid species, B_2O_3 , are treated at this time. Therefore, aqueous chemical reactions are not treated in the current model manifestation. Further studies are required to link the results from atomistic and thermodynamic simulations to calculate the most stable precipitating phase at any combination of CRUD and fluid conditions. The results in this paper also ignore boron burning for simplicity.

A baseline diffusion constant for boric acid of $1.07 \times 10^{-9} \frac{\text{m}^2}{\text{s}}$ at 298.15 K was taken from an EPRI report concerning their chemistry thermodynamics database [30]. It must therefore be scaled with temperature, which is done via ratios using the Stokes–Einstein

relation [31], finding a ratio between diffusion coefficients and temperatures:

$$\frac{D(T_1)}{D(T_2)} = \frac{T_1}{T_2} \cdot \frac{\mu(T_2)}{\mu(T_1)} \quad (34)$$

where $D(T)$ is the temperature-dependent diffusion constant, $\mu(T)$ is the temperature-dependent viscosity of water in Pa s, and T is the temperature in Kelvin. This relation is valid only for flows with a very low Reynolds number, which should hold for the slow fluid velocities in the CRUD.

The major sinks for boric acid are precipitation into boron oxide, and boron burning by an imposed neutron flux. These equilibrium data on ionic and solid mole fractions was acquired using the FactSage thermodynamic database. Sources are only likely to occur during a cooldown or shutdown, when the cooler fluid will re-dissolve retrograde boron-bearing compounds into their ionic forms. Precipitation will be discussed in detail below in Section 3.4.1.

It should be noted that evaporation at the chimney wall will tend to drive the soluble species up the concentration gradient in opposition to normal diffusion. A stable equilibrium will be reached (for a given set of conditions and operating history) once migration by diffusion balances migration by fluid convection. Boric acid vaporization will contribute to diffusion balancing out convective transport away from the boiling chimney, although this process is not yet implemented.

3.4.1. Precipitation and dissolution of B- and Li-bearing species

For now, it is assumed in MAMBA-BDM that any supersaturation of boric acid leads to the precipitation of B_2O_3 and filling the pores of the solid portion of the CRUD (i.e. not the vapor chimneys), with any excess BO_3^{-3} contributing to this mass. It is important to note that FactSage predicted a solid B_2O_3 concentration about 2000 times higher than that of solid $\text{Li}_2\text{B}_4\text{O}_7$ for typical PWR operating conditions ($T = 600 \text{ K}$, $P = 15 \text{ MPa}$, 2–3 ppm Li, 1200 ppm boric acid, $\text{pH} = 7.2\text{--}7.5$). This is likely due to both the stability of each compound and the far lower Li^+ concentration in the coolant (2–3 ppm) as compared to BO_3^{-3} (1000–2000 ppm). However, lithium has a higher affinity for the polyborate ions than hydrogen, suggesting that over time, much (if not all) of the boron oxide will convert to $\text{Li}_2\text{B}_4\text{O}_7$.

3.5. Code implementation

2D axially symmetric geometries are modeled by assigning relevant bulk and boundary equations to the mesh element blocks and side sets, respectively. The mesh uses h-adaptivity to refine the solution around areas that contribute more error to the result. It also coarsens the solution in areas that contribute little error, enabling optimization between computational time and accuracy.

Implementing the code involves direct computation of parameters over length scales that vary by many orders of magnitude. These range from enthalpies of vaporization (order of 10^6) to permeabilities (order of 10^{-15}). As a result the JFNK solver has difficulty in accurately tracking small but important parameters, which get lost in the 'floating point noise' of the computational system [32]. To mediate this problem, MAMBA-BDM introduces an artificial scaling capability. The default SI units for length (and all units derived from them) can be scaled by any amount, and all calculations are adjusted accordingly. This results in a more accurate solution, a much faster convergence time, and a considerable reduction in computational work. In order to correctly use scaling, one simply has to adjust the scaling factor in the input file. MAMBA-BDM shifts all calculations by this factor, by using *aprepro* to create algebraically symbolic, rather than static, input files. For example, to shift from meters to millimeters, one applies a scaling

factor of 0.001, and divides the mesh size by 0.001. The computational requirements for such simulations are drastically reduced, from 5 to 10 min using loosely-coupled simulations [13] to 8–30 s on one CPU using MAMBA-BDM.

4. Results

In this section, key results that highlight improvements between MAMBA-BDM and previous models are presented. First, spatial distributions of key variables show differences, such as a lower peak CRUD temperature and a higher maximum fluid velocity compared to past models. Second, applying MAMBA-BDM to CRUD with different boiling chimney spacings shows that heat convection in the fluid is non-negligible for tightly spaced chimneys, which is representative of real and experimentally-simulated CRUD [19]. Third, a comparison between calculated effective thermal conductivities in MAMBA-BDM, previous models, and related experiments show that the assumption of liquid-saturated CRUD is invalid.

4.1. Steady-state spatial distributions

MAMBA-BDM results for important spatial variable distributions in the CRUD surrounding one boiling chimney, using the control volume Ω in Fig. 2f, are shown in Fig. 2. The CRUD in this simulation was assumed to be 50 μm thick, boiling chimneys were 4 μm in diameter, chimney density was set at $2 \times 10^9 \frac{\text{chim}}{\text{m}^2}$ (chimney cores spaced 25.2 μm apart), the cladding heat flux was $1 \frac{\text{MW}}{\text{m}^2}$, the bulk coolant was 600 K and contained 1200 ppm boric acid with a typical heat transfer coefficient at the CRUD/coolant boundary of $12,000 \frac{\text{W}}{\text{m}^2 \text{K}}$ [33], and a skeletal solid CRUD porosity of 50% was used. Vapor chimney free area did not contribute to this porosity.

Fig. 2a shows the temperature distribution, which predictably peaks on the cladding away from the boiling chimney. This is because wick boiling is assumed to occur on the CRUD-chimney boundary, cooling the CRUD in that vicinity. Peak cladding temperatures computed in MAMBA-BDM add precision to macroscale CRUD codes, as the temperature is observed to vary by up to 6 K over a distance of a few μm on the cladding. Fig. 2b shows a contour plot of the fluid pressure in the CRUD. A pressure drop of 2.4 kPa was observed from coolant to cladding, giving the CRUD a permeability of $2.31 \times 10^{-15} \text{m}^2$, which is similar to fine sandstone [34]. Fluid velocities in Fig. 2c exceeded $11.5 \frac{\text{mm}}{\text{s}}$, especially near the cladding. This is due to wick boiling removing the heat generated by the fuel via evaporation, and the bottom

15–20 μm of CRUD provide sufficient surface area to remove most of this heat. Consequently, the fluid flows straight through the CRUD over much of its thickness (perpendicular to the pressure contours), until turning towards the chimney near the region of maximum boiling. Finally, the soluble boric acid in Fig. 2d reaches a concentration factor of 16.5, which is consistent with previous simulations [9,13]. Precipitation in Fig. 2e occurs wherever the non-equilibrium concentration of boric acid exceeds its FactSage-calculated solubility. Studies using MAMBA-BDM have shown that this precipitation region increases with increasing temperature, assuming all other variables are held constant. This is likely due to the retrograde solubility of the precipitating boron-bearing compounds.

4.2. The role of heat convection

The effects of adding a fluid convection term inside the CRUD, neglected in previous studies [10,13,9], showed increasing importance as the spacing between boiling chimneys was decreased. The first clue that fluid convection was required came from non-dimensional analysis of fluid flow in the CRUD. Computation of the Nusselt number (Nu) using the Siedel-Tate correlation [35] showed $\text{Nu} \cong 0.13$, which implies that conduction and convection within the pores of the CRUD are not orders of magnitude different in importance. Calculation of the Peclet number, taking the characteristic length as the CRUD thickness times the capillary tortuosity τ , showed that the convection term sometimes outweighed the conduction term in the fluid, especially in the region of highest boiling heat flux, and therefore fastest fluid flow. The average Péclet numbers throughout the CRUD, lying in the range 0.25–0.5, are comparable to the value of 0.46 found by Pan et al. [9].

Example curves showing the change in peak clad temperature with and without fluid convection for different chimney spacings are shown below in Fig. 3. The narrowly-spaced chimney curves in Fig. 3 shows the case for a chimney core-to-core spacing of $10 \mu\text{m}$ ($1 \times 10^{10} \frac{\text{chim}}{\text{m}^2}$), which is representative of some of the observations from the Westinghouse's WALT loop [19]. Here, the addition of fluid convection reduces the peak cladding temperature by 2 K, which gives rise to a non-negligible change in the corrosion rate (estimated at 10–15%). In contrast, the widely-spaced chimney curves in Fig. 3 shows the results for a wider chimney core-to-core spacing of 25 μm ($1.6 \times 10^9 \frac{\text{chim}}{\text{m}^2}$). The addition of fluid convection cools the CRUD by about 0.25 K, which is negligible. These results show that as the boiling chimneys in CRUD get more dense, coupled effects become more important, and therefore cannot be neglected.

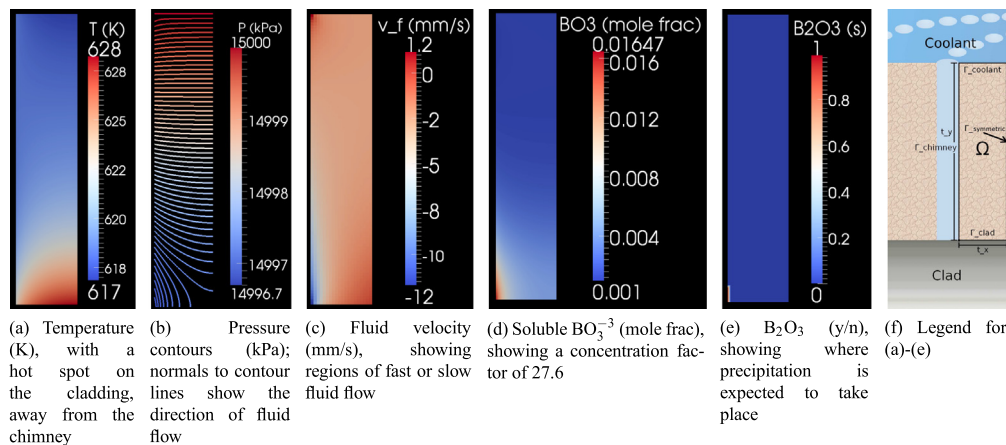


Fig. 2. Spatial distributions of main variables in the MAMBA-BDM microscale simulation in the region Ω in (f).

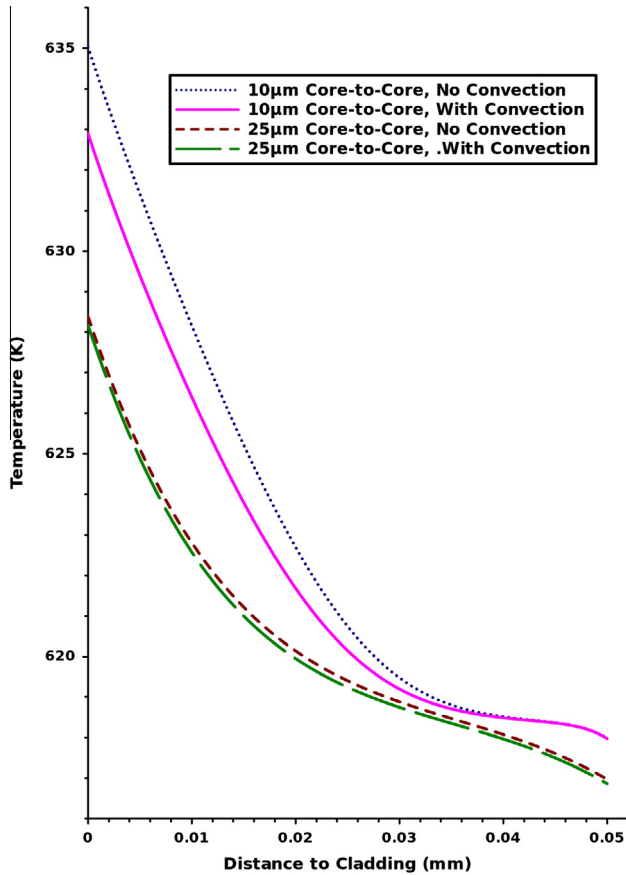


Fig. 3. Comparison of the effect of convection in CRUD for narrow (10 μm) and wide (25 μm) core-to-core chimney spacings.

4.3. The assumption of liquid saturation

In MAMBA-BDM, as in previous CRUD models, it was assumed that the fluid remained liquid while inside the CRUD. This is an artifact of the CRUD/chimney boundary condition, which imposes $T = T_{\text{sat}}$ on the CRUD-chimney wall. Analysis of the degree of superheating of the fluid in the CRUD is shown in Fig. 4. The non-dimensional analysis performed in Section 4.2 suggests that this amount of fluid superheating in Fig. 4 is not possible.

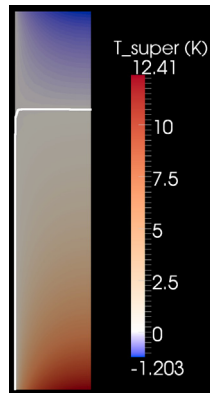


Fig. 4. Distribution showing the fluid superheat temperature ($T - T_{\text{sat}}$) for the simulation described in Section 4.1. The white line delineates the location of fluid saturation temperature.

A direct comparison of the calculated effective CRUD thermal conductivity in the WALT loop CRUD growth report [19] with that found in MAMBA-BDM is possible using the method of thermal resistances, and helps to explain the discrepancy in MAMBA-BDM vs. experimental observations. In the WALT loop, a single, vertical ohmically heated rod is immersed in an autoclave with PWR conditions, and CRUD precursors are added to the water. CRUD is then grown by sub-cooled boiling, similar to what happens in a PWR. Thermocouples are placed in the bulk coolant, and inside the inner diameter of the heated rod. The system is modeled using heat conduction through a hollow cylinder of CRUD, with the quantities r_{clad} , r_{CRUD} , $T_{\text{CRUD/clad}}$ & $T_{\text{CRUD/coolant}}$ defined as the cladding radius, the outer CRUD radius, the CRUD/clad temperature, and the CRUD/coolant temperature, respectively. The equation for heat flux through a cylindrical shell is used [36] in both our analysis and in Westinghouse's analysis of the WALT loop experiments [19]:

$$q'' = -\frac{k_{\text{effc}}}{r_{\text{clad}}} \frac{(T_{\text{CRUD/coolant}} - T_{\text{CRUD/clad}})}{\ln\left(\frac{r_{\text{CRUD}}}{r_{\text{clad}}}\right)} \quad (35)$$

where q'' is the surface heat flux produced by the fuel, taken at the outer diameter of the Zircaloy fuel cladding, and k_{effc} is the calculated effective thermal conductivity of the CRUD. This last term accounts for both conduction and boiling as heat removal mechanisms, combining them into one comparable parameter. Assuming that the cladding is 570 μm thick with an outer diameter of 9.80 mm ($r_{\text{clad}} = 5.27$ mm), and with simulated CRUD parameters as shown in Fig. 2a (50 μm thick $\rightarrow r_{\text{CRUD}} = 5.32$ mm, $T_{\text{CRUD/clad}} = 617.2$ K, $T_{\text{CRUD/coolant}} = 632.0$ K, $q'' = 1.5 \frac{\text{MW}}{\text{m}^2}$), an effective CRUD thermal conductivity of $5.04 \frac{\text{W}}{\text{mK}}$ is found. This is significantly higher than those measured or assumed in experimental [19,37] and theoretical [9,13] studies, respectively. A direct comparison of results is shown in Table 2, using the CRUD centerline temperature difference between the CRUD/clad interface and the CRUD/coolant interface. [9,13] both assumed thermal conductivities that match well with experimental values ($0.69 \frac{\text{W}}{\text{mK}}$ [9] and $0.506 \frac{\text{W}}{\text{mK}}$ [13]), but both models also assumed that the fluid in the CRUD's pores remained entirely liquid. This resulted in an effective thermal conductivity higher than that measured in experiments. The values of effective thermal conductivity calculated by MAMBA-BDM using assumptions similar to [9,13], however, are higher than both those calculated from previous models and those measured experimentally in [19,37]. All models that assume liquid saturation overestimate k_{effc} , and therefore overall heat transfer, through CRUD. Therefore, we believe that the addition of an additional CRUD dry-out region is necessary. This would provide a region of lower thermal conductivity in the vapor phase, as the thermal conductivity of dry CRUD is determined by the porosity and true thermal conductivity of its solid skeleton (neglecting the relatively small thermal conductivity of water vapor). The poor interconnectivity of the solid particles that comprise CRUD may also lower the thermal conductivity.

Table 2

Comparison of measured and calculated effective CRUD thermal conductivities with those in this study. Values for [9,13] were calculated using the minimum & maximum CRUD centerline temperatures and Eq. 35, assuming $r_{\text{clad}} = 5.27$ mm and taking all other parameters from the respective models.

Ref.	This work	Models		Experiments	
		[9]	[13]	[19] ^a	[37]
$k_{\text{effc}} \left[\frac{\text{W}}{\text{mK}} \right]$	5.04	4.40	2.29	0.519–1.39	0.82–1.82

^a Boiling cases (rods 110–117) used.

5. Discussion

5.1. Required changes to MAMBA-BDM based on results

As noted in Section 4.3, the assumption of liquid saturation highlights the shortcomings of assuming wick boiling in the CRUD. A more correct approach would be not to impose any temperature condition on $\Gamma_{chimney}$ at all, but to let it reach whatever temperature the physics allows. This does not restrict boiling to that face, and does not force boiling to occur along the entire chimney wall. The new pressure distribution must account for two-phase flow, but must also account for vapor transport through the boiling chimneys. This represents a momentum balance along the liquid/vapor path through the CRUD. The former Dirichlet boundary condition at the CRUD surface still applies:

$$P(\Gamma_{coolant}) = P_{coolant} \quad (36)$$

but now it applies to both liquid in the CRUD and vapor in the boiling chimney. In the liquid phase of the CRUD, Darcy flow may still apply, but with a new forcing function defined by capillary action of the pores at the liquid/vapor interface. This capillary pressure is defined as:

$$P_c = \frac{2\sigma \cos(\theta_w)}{\lambda/2} \quad (37)$$

where σ is the surface tension of the liquid, θ_w is its wetting angle, and λ is the average pore diameter in the CRUD.

5.2. Parallels between CRUD deposition and other phenomena

Of course, the formation of CRUD both influences and is influenced by the other phenomena occurring throughout the system. These interrelationships between the corrosion product deposition process and other parts of the system are highly analogous to many other non-nuclear systems, such as deposition in fossil fuel plants, fouling in heat exchangers, process heat delivery systems, and any other scenario involving at least solute transport in a temperature gradient.

The exploration of these highly coupled phenomena in CRUD therefore has wide reaching implications for similar, non-nuclear systems. The interrelationships explored and discussed between CRUD deposition, evolution, temperature, neutronics, and precipitation kinetics apply anywhere that some or all of these effects take place. This includes situations where a porous structure already exists, such as limestone deposits near a heat source, or situations where porous deposition takes place, like fouling in compact, shell-and-tube or printed circuit heat exchangers (PCHEs). The feedback effects between different physical mechanisms are too highly coupled to assume separate treatments, and only recently has computational power increased to the level that the average modeler can begin to simulate such complex systems. Positive feedback mechanisms, such as the exacerbation by precipitation on corrosion under CRUD and vice versa, are especially important to understand and quantify, as these have the most significant impact on accident or 'runaway' conditions that can lead to sudden, catastrophic failure of a system (for example, fuel rod leakage, or PCHE tube clogging).

Each and every coupling term must therefore be quantified and studied, both in terms of positive or negative feedback as well as determining the rate of each process. Such work has begun in the frame of MAMBA-BDM. Examples of areas of active study include: (1) Simulating simultaneous CRUD buildup and underlying cladding oxidation, (2) Incorporation of two-phase fluid models and capillary pressure, (3) Effects of coolant temperature drops or reactor shutdown on CRUD, (4) Coupling between radiolysis chemistry

and CRUD evolution, and (5) Feedback between CRUD surface temperature and heat removal as simulated by computational fluid dynamics.

6. Conclusions

MAMBA-BDM is being actively developed to track and predict peak cladding temperature and accumulation of boron and lithium bearing species (both soluble and insoluble) inside fuel CRUD over a range of time scales, from seconds to years. Coupled material models have been used, with preference given to those that can be directly verified by experimental results. Treating the interrelated phenomena involved with CRUD formation in a fully coupled manner has yielded new insights into how porous deposits with finitely soluble species behave, including the non-negligible effect of heat convection within. Comparing the detailed, 2D variable distributions within CRUD to experimental observations has rendered previous assumptions of liquid saturation in boiling deposits invalid under certain conditions, requiring a two-phase approach to be developed. Overall we have shown CRUD to be an appropriate prototypical nuclear materials challenge for multiscale modeling and simulation. The approach described here is applicable to other materials problems characterized by extreme environments and microstructure complexity at the mesoscopic scale.

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